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Glucokinase as glucose sensor and metabolic signal generator in pancreatic beta-cells and hepatocytes

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Abstract

This article reviews evidence for a pivotal role of glucokinase as glucose sensor of the pancreatic beta-cells. Glucokinase explains the capacity, hexose specificity, affinities, sigmoidicity, and anomeric preference of pancreatic islet glycolysis, and because stimulation of glucose metabolism is a prerequisite of glucose stimulation of insulin release, glucokinase also explains many characteristics of this beta-cell function. Glucokinase of the beta-cell is induced or activated by glucose in contrast to liver glucokinase, which is regulated by insulin. Tissue-specific regulation corresponds with observations that liver and pancreatic beta-cell glucokinase are structurally distinct. Glucokinase could play a glucose-sensor role in hepatocytes as well, and certain forms of diabetes mellitus might be due to glucokinase deficiencies in pancreatic beta-cells, hepatocytes, or both.

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